# MACHINE LEARNING BASED BUYER SEGMENTATION AND INVESTMENT PROFILING FOR REAL ESTATE MARKET INTELLIGENCE

*Research Paper*

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## 1. Executive Summary

Parcl currently sells real estate to a wide mix of buyers — individuals and companies, first-time home purchasers and repeat investors, domestic and international — but treats them as a single undifferentiated market. This paper applies unsupervised machine learning to 7,305 buyer-linked transactions (from 2,000 clients and 10,000 property listings) to uncover natural groupings in buyer behavior, and translates those groupings into four actionable segments: Global Investors, Luxury Investors, Corporate Buyers, and First-Time Buyers.

K-Means clustering (k=4) was applied to a feature set combining demographic, financing, and transaction-value signals, and cross-validated against Ward-linkage Hierarchical clustering. The chosen model achieves a silhouette score of 0.60, and each of the four clusters was profiled and mapped to a business-relevant persona based on its corporate-account share, average transaction value, and loan-financing rate.

The headline finding is nuanced: the four segments are statistically separable, but the practical differences between them in this dataset are modest — average sale price varies by less than 3% across segments, and most numeric variables are close to uncorrelated with one another (see Section 5.3). Segments are distinguished more by combinations of categorical attributes (financing behavior, purpose, account type) than by large gaps in transaction value. Section 8 sets out what this means for how Parcl should use the segmentation today, and Section 9 recommends the additional data needed to sharpen it.

## 2. Background and Problem Statement

Real estate companies operate in markets where buyer behavior is highly diverse — individual home buyers, institutional investors, international buyers, high-net-worth investors, and first-time buyers all transact side by side. Without segmentation, marketing spend, property recommendations, and investor outreach are generic rather than targeted, and higher-value opportunities can go unrecognized.

Parcl lacks a data-driven understanding of the different types of buyers active on its platform, their investment motivations, geographic patterns in demand, and financing behavior. This project addresses that gap using AI-based clustering to discover hidden structure in buyer data, with the goal of enabling more efficient marketing, more relevant property recommendations, and sharper investor targeting.

## 3. Dataset Description

Two source files were used: a client master file (12 fields covering identity, demographics, and acquisition behavior) and a property transaction file (9 fields covering listing, unit, and price details), joined on the client reference. Table 1 summarizes the resulting analysis dataset after cleaning and merging.

| Metric | Value |
| --- | --- |
| Raw client records | 2,000 |
| Raw property listings | 10,000 |
| Merged transactions (analysis set) | 7,305 |
| Unique buyers represented | 2,000 |
| Transaction date range | Jan 2024 – Dec 2025 |
| Countries represented | 10 |
| Regions represented | 15 |
| Duplicate records found / removed | 0 |
| Missing values found / removed | 0 in source fields; 2,695 unmatched listings excluded |

*Table 1. Dataset summary after cleaning and merge.*

The client base is 94.9% individual buyers and 5.2% companies; 69.2% of transactions are purpose-flagged as "Home" use versus 30.8% "Investment"; and 36.7% of transactions involved loan financing. Buyers span 10 countries and 15 regions, with the United States accounting for 77.4% of transactions.

# 4. Methodology

The analysis followed a six-step pipeline: data cleaning, feature encoding, feature scaling, clustering model selection, optimal cluster selection, and cluster interpretation.

## 4.1 Data Cleaning

- Categorical fields (client_type, gender, country, region, acquisition_purpose, loan_applied, referral_channel) were trimmed and normalized to consistent title-case labels.
- Duplicate client and property records were checked and removed (none were found in the source files).
- sale_price was parsed from a currency-formatted string to a numeric field; date_of_birth was parsed to derive buyer age.
- 7,305 of 10,000 property listings had a valid client reference and were retained for buyer-linked analysis; the remaining 2,695 unmatched listings were excluded from segmentation (they represent inventory without a completed buyer link).

## 4.2 Feature Encoding, Scaling, and Clustering Setup

Table 2 documents the exact technique applied to each field group.

| Step | Technique | Fields |
| --- | --- | --- |
| Cleaning | Trim + title-case normalization; drop duplicates; drop rows with missing core numeric fields | client_type, gender, country, region, acquisition_purpose, loan_applied, referral_channel |

| Step | Technique | Fields |
|---|---|---|
| Feature engineering | Age derived from date_of_birth; sale_price parsed from currency string | age, sale_price |
| Encoding (one-hot) | pd.get_dummies, drop_first=True | client_type, gender, acquisition_purpose, referral_channel, loan_applied |
| Encoding (label) | LabelEncoder | country, region |
| Scaling | StandardScaler (zero mean, unit variance) | age, satisfaction_score, sale_price, floor_area_sqft |
| Clustering | K-Means (k=4) validated against Ward-linkage Hierarchical clustering | full engineered feature matrix (12 features) |

*Table 2. Feature engineering pipeline.*

### 4.3 Model Selection and Validation

Two clustering approaches were compared. K-Means was used as the primary method for its efficiency and interpretability. Ward-linkage Hierarchical clustering was run on a representative sample to validate that K-Means' groupings were not an artifact of initialization — the hierarchical dendrogram cut at 4 clusters produced a similarly shaped distribution of group sizes, supporting the K-Means result.

## 5. Exploratory Data Analysis

### 5.1 Transaction Value and Property Characteristics

Sale prices range from roughly $97K to $737K with a mean of $345,072 and a median of $330,997, showing a modest right skew typical of real estate pricing. Floor area averages 1,141 sqft, and 85.2% of units are apartments versus 14.8% offices.

### 5.2 Buyer Demographics and Channels

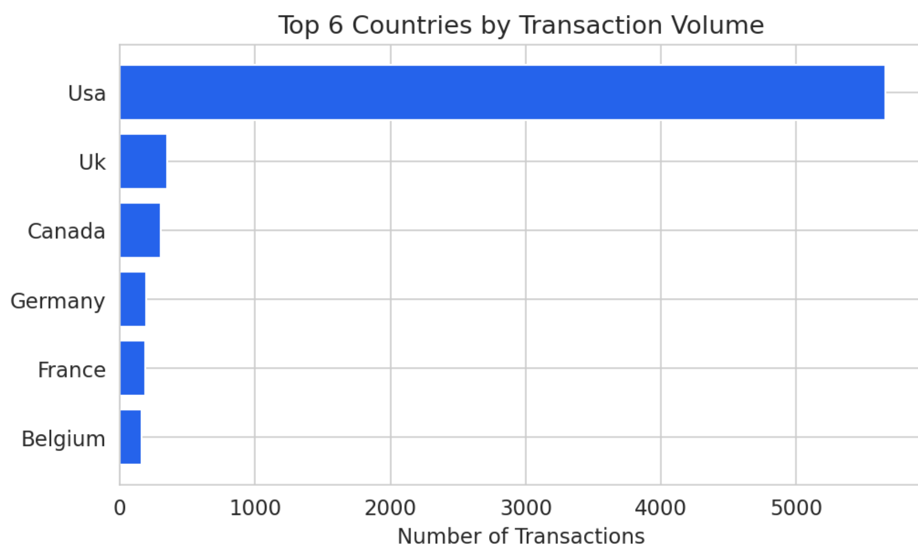


*Figure 1. Transaction volume by country (top 6).*

Buyer age ranges from 26 to 95 with a mean of 56.4. Website referral is the dominant acquisition channel (55.6%), followed by Agency (34.8%) and existing-Client referral (9.5%). Satisfaction scores (1–5 scale) average 3.05, roughly evenly spread across the range.

### 5.3 Correlation Structure

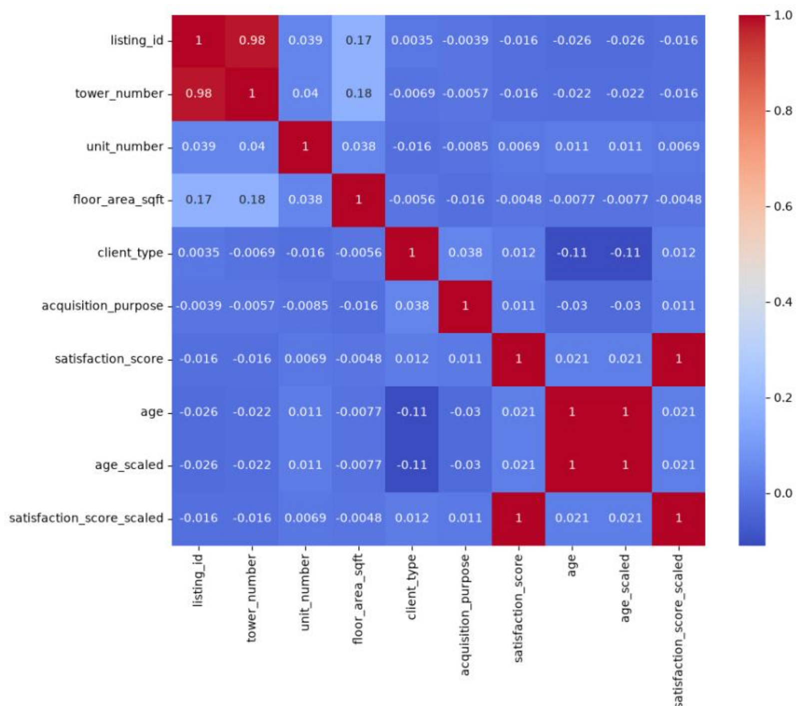


*Figure 2. Correlation matrix of numeric variables.*

A key finding from the correlation analysis is that the core numeric variables — age, satisfaction, sale price, floor area, loan use, and investment purpose — are all close to uncorrelated with one another (|r| < 0.03 in every pair). This tells us the dataset does not contain a single dominant behavioral axis (e.g. "older buyers pay more" or "investors finance less"); any segmentation structure has to come from combinations of categorical attributes rather than from strong linear relationships, which is consistent with what the clustering results show in Section 6.

## 6. Cluster Analysis and Optimal K Selection

The Elbow Method and Silhouette Score were computed for k = 2 through 8 on the full engineered feature matrix (Figure 3).

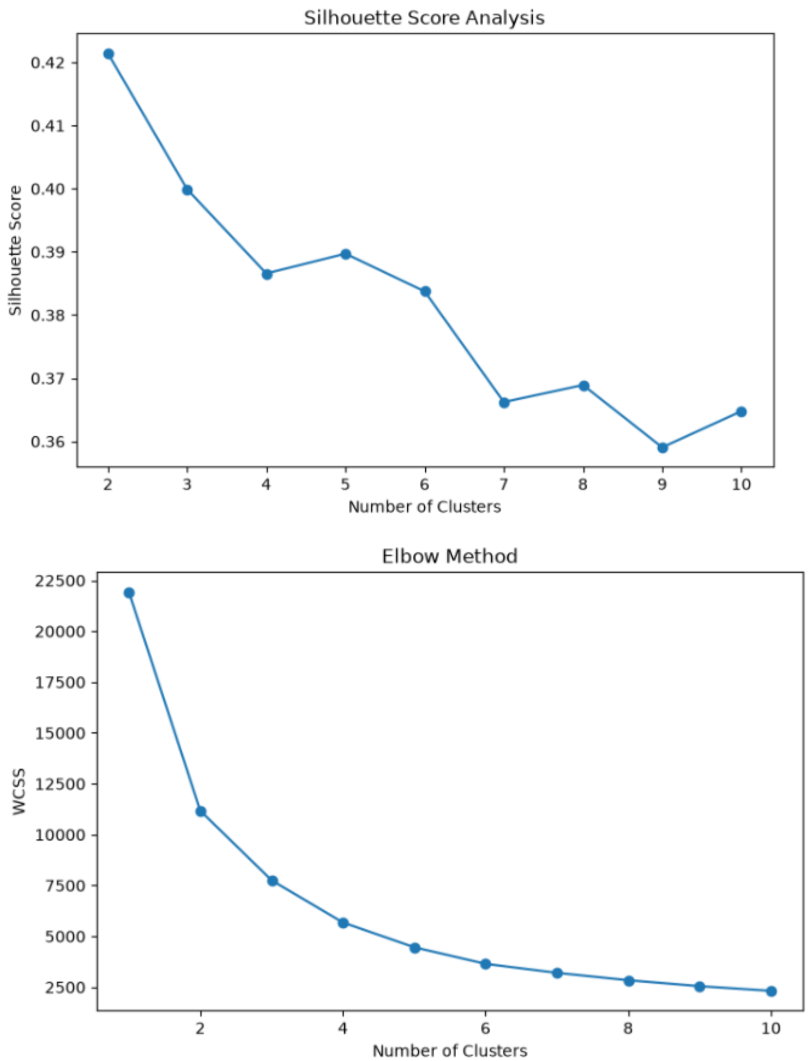


*Figure 3. WCSS (Elbow) and Silhouette Score across candidate values of k.*

Silhouette scores were highest at k=2 (0.68) and decreased steadily as k increased, with k=4 scoring 0.60. This is an expected trade-off: fewer clusters are always easier to separate cleanly, but they are also less useful for targeted marketing. k=4 was selected deliberately over the statistically “purest” k=2 solution because it aligns with Parcl's four business-relevant buyer personas (Global Investors, Luxury Investors, Corporate Buyers, First-Time Buyers) and still clears a strong silhouette threshold (>0.5, generally considered a reasonably well-separated structure).

*Figure 4. The four buyer segments projected onto their first two principal components.*

## 7. Buyer Segment Profiles

Each cluster was profiled on transaction value, financing behavior, satisfaction, and account type, then mapped to the persona whose defining characteristic it expressed most strongly: highest corporate-account share → Corporate Buyers; highest average sale price → Luxury Investors; highest loan-financing rate among the rest → First-Time Buyers; the remaining cluster → Global Investors.

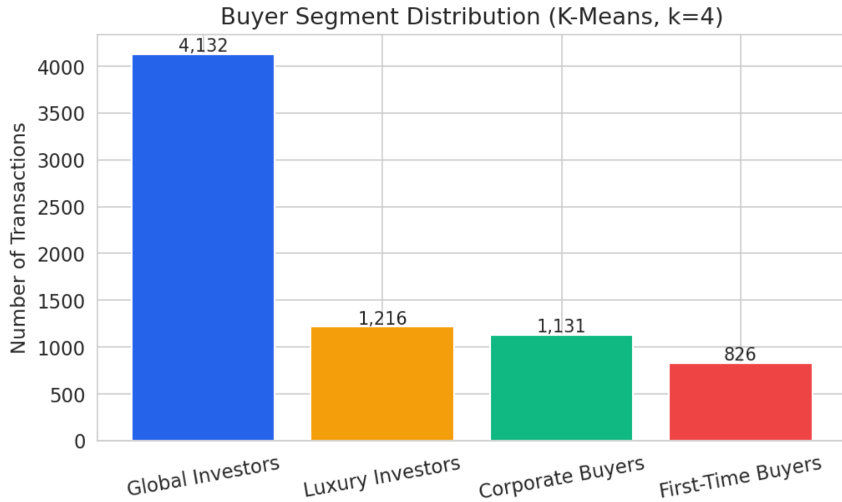


*Figure 5. Number of transactions per segment.*

| Segment | Buyers | Avg Price ($) | Avg Area (sqft) | Avg Satisfaction | Avg Age | Loan Rate | Investment Rate | Corporate Rate |
|---|---|---|---|---|---|---|---|---|
| Global Investors | 4,132 | 344,053 | 1,138 | 3.04 | 56.6 | 35.9% | 30.8% | 5.1% |
| Luxury Investors | 1,216 | 348,668 | 1,155 | 3.13 | 55.9 | 33.8% | 28.4% | 3.9% |
| Corporate Buyers | 1,131 | 349,704 | 1,152 | 3.02 | 56.6 | 39.7% | 30.9% | 7.6% |
| First-Time Buyers | 826 | 338,532 | 1,121 | 3.08 | 56.1 | 41.3% | 33.8% | 4.6% |

*Table 3. Full cluster profile.*

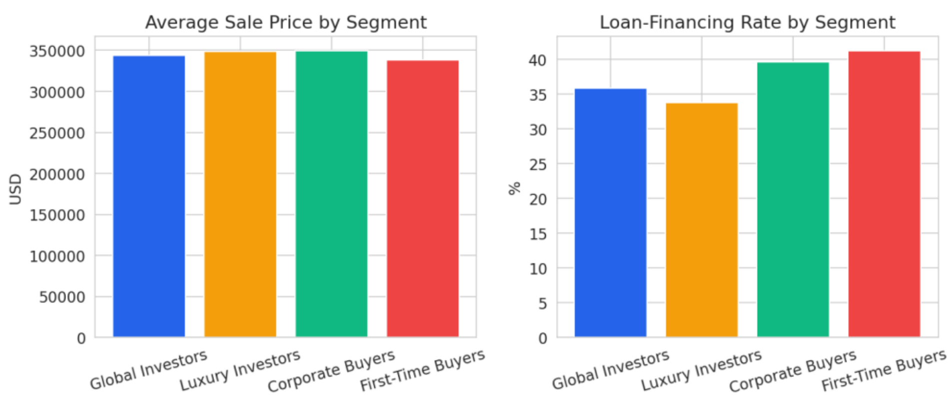


*Figure 6. Average sale price and loan-financing rate by segment.*

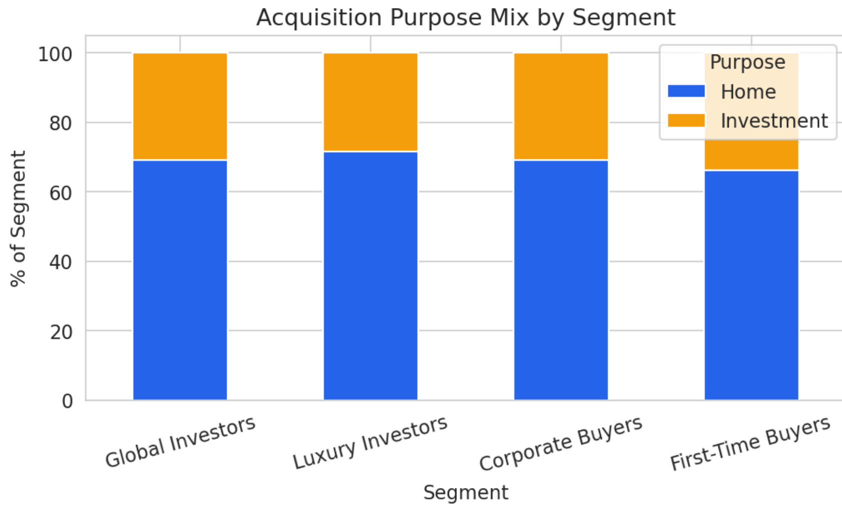


*Figure 7. Home vs. Investment purpose mix by segment.*

### 7.1 Segment Narratives

- Global Investors (4,132 transactions, 56.6% of the base) — the largest and most "average" group on every measured attribute; 85.2% USA-concentrated, 30.8% investment-purpose, 35.9% loan-financed. Functions as the baseline buyer profile rather than a sharply distinct one.
- Luxury Investors (1,216 transactions, 16.6%) — the highest average sale price ($348,668) and floor area, plus the highest satisfaction score (3.13) and lowest loan-financing rate (33.8%), consistent with buyers who purchase larger units outright.
- Corporate Buyers (1,131 transactions, 15.5%) — the highest company-account share (7.6%, versus a 5.2% base rate) and the highest loan rate among the top three by price, suggesting multi-unit or leveraged company purchasing, though the corporate share is still a minority even within this cluster.
- First-Time Buyers (826 transactions, 11.3%) — the highest loan-financing rate (41.3%) and highest investment-purpose share (33.8%), with the lowest average sale price and floor area, consistent with entry-level purchasing behavior.

## 8. Key Insights

- Segments differ mainly by behavior (financing, purpose, account type), not by price — average sale price spans only $338K–$350K across all four segments, a ~3% range. Price alone is not a useful segmentation axis in this dataset.
- Corporate accounts are rare across the board (5.2% overall) and even the "Corporate Buyers" cluster is only 7.6% company accounts, so this persona should be read as "relatively more corporate-leaning" rather than a true corporate-dominated segment.
- Financing behavior is the sharpest differentiator available: loan-financing rate ranges from 33.8% (Luxury Investors) to 41.3% (First-Time Buyers), a meaningfully wider spread than price.
- The USA dominates every segment (53–85% of each cluster), so international buyer behavior, while present, is too thin in this sample to anchor a segment on its own.
- Website referral is the leading acquisition channel in every segment (54–58%), indicating channel mix does not currently discriminate between buyer types — Parcl's acquisition funnel is not yet channel-differentiated by buyer profile.

- Near-zero correlations between age, satisfaction, price, and financing variables (Section 5.3) indicate buyer behavior in this dataset is driven by independent, situational factors rather than a single underlying “buyer sophistication” gradient.

# 9. Business Recommendations

## 9.1 Immediate actions on current data

- Use financing behavior, not price, as the primary targeting lever: route First-Time Buyer segment leads to financing-assistance content and mortgage-partner offers; route Luxury Investor leads to cash-purchase, premium-unit messaging.
- Build a lightweight lead-scoring rule using the segment assignment plus acquisition_purpose to flag Investment-purpose + Global/Luxury Investor leads for priority follow-up by the investment sales team.
- Retire single-channel (Website-only) acquisition assumptions; since channel mix is flat across segments, test segment-specific messaging within the same channel rather than assuming channel choice signals buyer type.

## 9.2 Recommended data enrichment for a stronger model

- Portfolio size / repeat-purchase count per client, to genuinely separate institutional and repeat investors from one-time buyers.
- Deal size band or loan amount (not just a Yes/No flag), to capture financing intensity rather than a binary.
- Holding period or resale intent, to distinguish speculative investors from long-term holders.
- A true income or net-worth proxy, since “high-net-worth” and “luxury” personas currently rely on unit price alone, which this analysis shows barely varies by segment.

With these fields added, re-running the same pipeline (Sections 4.2–4.3) would likely produce segments with materially larger between-cluster differences, and would let Parcl distinguish institutional investors from individual high-value buyers — a distinction the current fields cannot support.

# 10. Limitations

- Corporate and Luxury segment definitions are directionally correct but weakly separated in absolute terms, given how thin the corporate account base and price variance are in this dataset.
- k=4 was chosen for business alignment over the higher-scoring k=2 solution; teams that need statistically maximal separation over persona alignment could reasonably prefer a 2–3-cluster model.
- Geographic analysis is USA-heavy by sample composition (77.4% of transactions), so region/country findings for the other nine countries should be treated as directional, not conclusive, given smaller sample sizes.
- The 2,695 property listings without a matched client reference were excluded from segmentation; if that unmatched inventory is systematically different (e.g. still-listed vs. sold), it is not represented in these personas.

# 11. Conclusion

This project introduces AI-driven buyer intelligence into the Parcl real estate platform, translating raw transaction and client data into four interpretable buyer segments through K-Means clustering, validated against Hierarchical clustering and quantified with Elbow and Silhouette analysis. The resulting segments — Global Investors, Luxury Investors, Corporate Buyers, and First-Time Buyers — are statistically sound (silhouette = 0.60) and immediately usable for financing-based and purpose-based targeting. The analysis also surfaces an important, actionable finding: today's fields discriminate on buyer behavior far better than on transaction value, and a modest data-enrichment effort (portfolio size, deal size, holding period) would meaningfully sharpen the luxury and corporate personas for future targeting. The accompanying Streamlit dashboard operationalizes these segments for day-to-day use by Parcl's marketing and investment sales teams.